# NATGRID Hybrid Encryption - Admin Guide

## 1. What is Hybrid Encryption?

Hybrid Encryption combines Symmetric Encryption (AES) and Asymmetric Encryption (RSA). AES encrypts the actual data because it is fast, while RSA encrypts the AES key so that only the receiver can decrypt it.

## 2. Algorithms Used

- AES-256-CBC: Encrypts the data file.
- RSA Public/Private Key: Encrypts the AES key.
- X.509 Certificate: Used to extract the NATGRID public key.

## 3. Data Encryption Workflow(PO's)

Receive NATGRID SSL certificate.

Extract NATGRID public key:
openssl x509 -pubkey -noout -in certificate.crt -out natgrid_public.pem

Generate AES key:
openssl rand -hex 32 > aes_key

Generate IV:
openssl rand -hex 16 > aes_iv

Encrypt data:
AES_KEY=$(cat aes_key)
AES_IV=$(cat aes_iv)
openssl enc -aes-256-cbc -K $AES_KEY -iv $AES_IV -in data.file -out data.file.enc

Encrypt AES key using NATGRID public key:
openssl rsautl -encrypt -pubin -inkey natgrid_public.pem -in aes_key -out aes_key.enc

**Note: Recive this (data.file.enc, aes_key.enc and aes_iv) files from Users.**

## 4. Data Decryption Workflow(NATGRID)

Decrypt AES key:
openssl rsautl -decrypt -inkey natgrid_private.pem -in aes_key.enc -out aes_key

Read AES key and IV.

Decrypt file:
AES_KEY=$(cat aes_key)
AES_IV=$(cat aes_iv)
openssl enc -d -aes-256-cbc -K $AES_KEY -iv $AES_IV -in data.file.enc -out data.file

**Note: Verify decrypted file.**

## 5. Architecture

Airtel -> Generate AES Key -> Encrypt Data (AES) -> Encrypt AES Key (RSA Public Key) -> Send Encrypted File + Encrypted AES Key + IV -> NATGRID -> Decrypt AES Key (RSA Private Key) -> Decrypt Data (AES).

## 6. Why Hybrid Encryption?

AES is very fast for large files. RSA is secure for key exchange. Combining both provides high performance and strong security.